Advanced Statistical Methods II
Lecture IV: EM Algorithm — First Principles

Monitirtha Dey

BSDS Third Year Students
Indian Statistical Institute

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Outline of Today's Lecture

Where we are in the course

Imputation: "filling in values". using the conditional distribution of missing quantities given observed information to construct plausible completed data.

Today's focus: likelihood-based estimation with missing or latent data.

- The complete-data likelihood may be simple.
- The observed-data likelihood may require summing or integrating over what we did not observe.
- EM turns that difficult optimization problem into a sequence of easier conditional-expectation and maximization steps.

Expectation $\longrightarrow$ Maximization $\longrightarrow$ Repeat

A brief history of the EM algorithm

The idea of alternating between estimating missing information and updating parameters appeared in several earlier special cases, especially in incomplete-data likelihood problems.

The modern general formulation was given by Dempster–Laird–Rubin in their seminal paper Dempster et al. (1977).

- It introduced the name **EM algorithm**: **E**xpectation followed by **M**aximization.
- It showed that many apparently different procedures could be understood through one incomplete-data likelihood framework. It turned a collection of problem-specific tricks into a general statistical algorithm.
- The paper also established the fundamental likelihood-ascent idea that makes EM statistically attractive.

See historical comments in McLachlan and Krishnan (2008) Ch. 1.

A simple motivating example

Suppose items are produced by one of two machines, but for some reason the machine label is not recorded.

For item i:

$$Y_i = \text{number of defects}, \qquad Z_i = \begin{cases} 1, & \text{Machine 1}, \\ 2, & \text{Machine 2}. \end{cases}$$

Assume

$$\mathbb{P}(Z_i = 1) = \pi, \qquad Y_i \mid Z_i = k \sim \text{Poisson}(\lambda_k), \quad k = 1, 2.$$

We observe only

$$y_1, \ldots, y_n,$$

but not $z_1, \ldots, z_n$.

Can we estimate $\pi, \lambda_1, \lambda_2$ without knowing which machine produced which item?

Why this example is an EM problem

If the machine labels Z_i's were known, estimation would be easy.

But they are not known, so the likelihood contribution of observation i is

$$\pi \frac{e^{-\lambda_1}\lambda_1^{y_i}}{y_i!} + (1-\pi)\frac{e^{-\lambda_2}\lambda_2^{y_i}}{y_i!}.$$

Hence the observed-data log-likelihood is

$$\ell(\pi, \lambda_1, \lambda_2) = \sum_{i=1}^{n} \log\left\{\pi f_1(y_i; \lambda_1) + (1-\pi) f_2(y_i; \lambda_2)\right\}.$$

The difficulty is the *log of a sum*. It prevents the likelihood from separating into simple machine-specific pieces.

Complete-data version: what would be easy?

Let

$$Z_{i1} = \mathbb{I}(Z_i = 1), \qquad Z_{i2} = \mathbb{I}(Z_i = 2) = 1 - Z_{i1}.$$

If these indicators were observed, the complete-data log-likelihood would be, up to constants,

$$\ell_c(\theta) = \sum_{i=1}^{n} \left[Z_{i1}\{\log \pi + y_i \log \lambda_1 - \lambda_1\} + Z_{i2}\{\log(1-\pi) + y_i \log \lambda_2 - \lambda_2\} \right],$$

where $\theta = (\pi, \lambda_1, \lambda_2)$.

With the labels known, the problem separates into two ordinary Poisson MLE problems plus an estimate of the mixing proportion.

Deriving the complete-data update for π

Focus on the terms in ℓ_c involving π:

$$\ell_c(\pi) = \sum_{i=1}^{n} \{ Z_{i1} \log \pi + Z_{i2} \log(1-\pi) \} + C.$$

Differentiate:

$$\frac{\partial \ell_c}{\partial \pi} = \sum_{i=1}^{n} \frac{Z_{i1}}{\pi} - \sum_{i=1}^{n} \frac{Z_{i2}}{1-\pi}.$$

Set the derivative equal to zero:

$$\frac{\sum_i Z_{i1}}{\pi} = \frac{\sum_i Z_{i2}}{1-\pi}.$$

Since $Z_{i1} + Z_{i2} = 1$, this gives

$$\widehat{\pi} = \frac{1}{n} \sum_{i=1}^{n} Z_{i1}.$$

Deriving the complete-data updates for λ_1, λ_2

For λ_1, keep only the relevant terms:

$$\ell_c(\lambda_1) = \sum_{i=1}^{n} Z_{i1}\{y_i \log \lambda_1 - \lambda_1\} + C.$$

Differentiate and set equal to zero:

$$\frac{\partial \ell_c}{\partial \lambda_1} = \sum_{i=1}^{n} Z_{i1}\left(\frac{y_i}{\lambda_1} - 1\right) = 0.$$

Thus

$$\widehat{\lambda}_1 = \frac{\sum_i Z_{i1} y_i}{\sum_i Z_{i1}}.$$

Similarly,

$$\widehat{\lambda}_2 = \frac{\sum_i Z_{i2} y_i}{\sum_i Z_{i2}}.$$

Complete-data MLEs if labels were known

From the complete-data log-likelihood, the MLEs would be

$$\widehat{\pi} = \frac{1}{n}\sum_{i=1}^{n} Z_{i1},$$

$$\widehat{\lambda}_1 = \frac{\sum_{i=1}^{n} Z_{i1} y_i}{\sum_{i=1}^{n} Z_{i1}}, \qquad \widehat{\lambda}_2 = \frac{\sum_{i=1}^{n} Z_{i2} y_i}{\sum_{i=1}^{n} Z_{i2}}.$$

So the only obstacle is that the Z's are missing.

EM handles this by (in its M step) replacing the missing indicators (Z_{ik}) by their conditional expectations and then use the revised formulas.

E-step for the motivating example

At iteration t, suppose the current values are

$$\theta^{(t)} = (\pi^{(t)}, \lambda_1^{(t)}, \lambda_2^{(t)}).$$

Compute the posterior probability that observation i came from Machine 1:

$$\tau_i^{(t)} = \mathbb{P}(Z_i = 1 \mid Y_i = y_i, \theta^{(t)}) = \frac{\pi^{(t)} f_1(y_i; \lambda_1^{(t)})}{\pi^{(t)} f_1(y_i; \lambda_1^{(t)}) + (1 - \pi^{(t)}) f_2(y_i; \lambda_2^{(t)})}.$$

Then

$$\mathbb{E}[Z_{i1} \mid y_i, \theta^{(t)}] = \tau_i^{(t)}, \qquad \mathbb{E}[Z_{i2} \mid y_i, \theta^{(t)}] = 1 - \tau_i^{(t)}.$$

M-step for the motivating example

The M-step uses the complete-data MLE formulas, but replaces missing labels by responsibilities.

Thus

$$\pi^{(t+1)} = \frac{1}{n} \sum_{i=1}^{n} \tau_i^{(t)},$$

$$\lambda_1^{(t+1)} = \frac{\sum_{i=1}^{n} \tau_i^{(t)} y_i}{\sum_{i=1}^{n} \tau_i^{(t)}}, \qquad \lambda_2^{(t+1)} = \frac{\sum_{i=1}^{n} (1 - \tau_i^{(t)}) y_i}{\sum_{i=1}^{n} (1 - \tau_i^{(t)})}.$$

The update is exactly what we would do with known labels, except that each observation receives fractional membership in the two machines.

One numerical EM iteration

Suppose the observed defect counts are

$$0, \quad 1, \quad 2, \quad 5, \quad 6,$$

and start with

$$\pi^{(0)} = 0.5, \qquad \lambda_1^{(0)} = 1, \qquad \lambda_2^{(0)} = 5.$$

The E-step gives approximately:

y_i	0	1	2	5	6
$\tau_i^{(0)}$	0.982	0.916	0.686	0.017	0.003

Small counts are assigned mostly to Machine 1; large counts are assigned mostly to Machine 2.

Completing the numerical update

Using the values from the previous slide,

$$\sum_i \tau_i^{(0)} \approx 2.605, \qquad \sum_i \tau_i^{(0)} y_i \approx 2.395.$$

Therefore

$$\pi^{(1)} \approx \frac{2.605}{5} = 0.521,$$

$$\lambda_1^{(1)} \approx \frac{2.395}{2.605} = 0.919, \qquad \lambda_2^{(1)} \approx 4.845.$$

Continuing the motivating example

Starting from

$$(\pi^{(0)}, \lambda_1^{(0)}, \lambda_2^{(0)}) = (0.500, 1.000, 5.000),$$

one EM iteration gave

$$(\pi^{(1)}, \lambda_1^{(1)}, \lambda_2^{(1)}) \approx (0.521, 0.919, 4.845).$$

Repeating the same E-step and M-step once more gives

$$(\pi^{(2)}, \lambda_1^{(2)}, \lambda_2^{(2)}) \approx (0.515, 0.903, 4.816),$$

and a third iteration gives

$$(\pi^{(3)}, \lambda_1^{(3)}, \lambda_2^{(3)}) \approx (0.511, 0.894, 4.793).$$

Estimates and likelihood values over iterations

For this example, the observed-data log-likelihood is

$$\ell(\pi, \lambda_1, \lambda_2) = \sum_{i=1}^{5} \log \left\{ \pi f(y_i; \lambda_1) + (1 - \pi) f(y_i; \lambda_2) \right\}.$$

Iteration	π	λ_1	λ_2	$\ell(\theta)$
0, start	0.500	1.000	5.000	-10.318
1	0.521	0.919	4.845	-10.298
2	0.515	0.903	4.816	-10.296
3	0.511	0.894	4.793	-10.295

The log-likelihood becomes less negative, hence larger, after each EM iteration shown here.

What the motivating example teaches

This example already contains the full EM logic:

1. The observed likelihood is hard because latent labels have been summed out.
2. The complete-data likelihood would be easy if the labels were known.
3. The E-step computes conditional expectations of the missing labels.
4. The M-step maximizes the expected complete-data log-likelihood.
5. Repeating the two steps produces a sequence of likelihood-improving estimates.

The rest of the lecture formalizes this pattern.

Finite-mixture EM examples are treated in McLachlan and Krishnan (2008), Ch. 1.

Complete-data versus observed-data likelihood

Let Z denote all unobserved or latent quantities.

The complete-data likelihood is

$$L_c(\theta; y, z) = p(y, z \mid \theta).$$

The observed-data likelihood is obtained by eliminating Z:

$$L(\theta; y) = p(y \mid \theta) = \sum_z p(y, z \mid \theta)$$

for discrete Z, or

$$L(\theta; y) = \int p(y, z \mid \theta)\, dz$$

for continuous Z.

See Little and Rubin (2019), Ch. 8; McLachlan and Krishnan (2008), Ch. 1.

The difficulty is computational and algebraic:

- **Marginalization creates sums or integrals.** These may not simplify analytically.
- **A log of a sum is awkward.** Even if $p(y, z \mid \theta)$ factors nicely, $\log \sum_z p(y, z \mid \theta)$ usually does not inherit that simple factorization.
- **Direct derivatives may become nonlinear.** Closed-form MLEs can disappear even when complete-data MLEs are easy.

EM does not change the likelihood we want to maximize. It changes *how* we maximize it.

The EM idea in one sentence

Given a current parameter value $\theta^{(t)}$, replace the missing complete-data information by its conditional expectation under $\theta^{(t)}$, then maximize the resulting expected complete-data log-likelihood to obtain $\theta^{(t+1)}$.

Symbolically:

$$\theta^{(t)} \xrightarrow{\text{E-step}} Q(\theta \mid \theta^{(t)}) \xrightarrow{\text{M-step}} \theta^{(t+1)}.$$

This is repeated until the parameter estimates or likelihood values stabilize.

Notation for the algorithm

We write:

- Y: observed data,
- Z: missing or latent data,
- θ: parameter vector,
- $\ell_c(\theta; Y, Z) = \log p(Y, Z \mid \theta)$: complete-data log-likelihood,
- $\ell(\theta; Y) = \log p(Y \mid \theta)$: observed-data log-likelihood.

At iteration t, the current estimate is $\theta^{(t)}$.

The distinction between ℓ_c and ℓ is essential: EM works with the first to improve the second.

The central object: the Q-function

Define

$$Q(\theta \mid \theta^{(t)}) = \mathbb{E}_{\theta^{(t)}}[\ell_c(\theta; Y, Z) \mid Y] .$$

Interpretation:

- Treat θ inside ℓ_c as the parameter we want to optimize.
- Use the current value $\theta^{(t)}$ to determine the conditional distribution of $Z \mid Y$.
- Average the complete-data log-likelihood over that conditional distribution.

The two appearances of θ play different roles: θ is the candidate parameter; $\theta^{(t)}$ defines the expectation.

E-step: what exactly is computed?

Expectation step:

$$Q(\theta \mid \theta^{(t)}) = \mathbb{E}_{\theta^{(t)}}[\ell_c(\theta; Y, Z) \mid Y].$$

In practice, this often reduces to computing conditional expectations such as

$$\mathbb{E}[Z_i \mid Y, \theta^{(t)}], \qquad \mathbb{E}[Z_i^2 \mid Y, \theta^{(t)}],$$

or posterior class probabilities

$$\mathbb{P}(Z_i = k \mid Y_i, \theta^{(t)}).$$

The E-step does *not* necessarily fill in one value for each missing quantity. It averages over uncertainty in the missing quantities.

M-step: what exactly is maximized?

Maximization step:

$$\theta^{(t+1)} = \arg\max_{\theta} Q(\theta \mid \theta^{(t)}).$$

The M-step treats the expectations computed in the E-step as fixed quantities.

- If the complete-data MLE has a closed form, the M-step often inherits a simple update.
- If exact maximization is difficult, a generalized EM step may merely increase Q; we will discuss that later in the course.

EM and generalized EM: Dempster et al. (1977); McLachlan and Krishnan (2008).

The basic EM algorithm

Algorithm

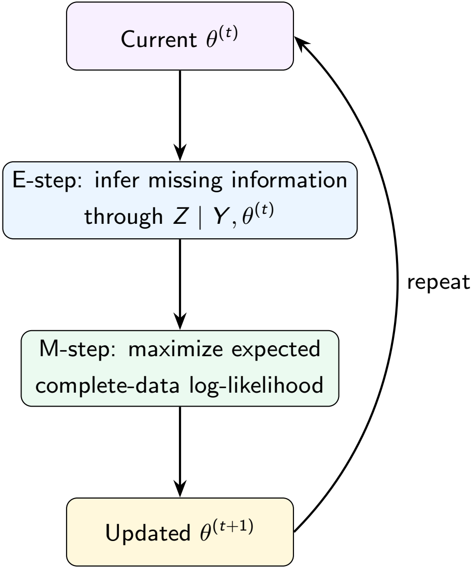
1. Choose a starting value $\theta^{(0)}$.
2. For $t = 0, 1, 2, \ldots$:
   - E: Compute $Q(\theta \mid \theta^{(t)})$.
   - M: Set
   $$\theta^{(t+1)} = \arg\max_{\theta} Q(\theta \mid \theta^{(t)}).$$
3. Stop when a convergence criterion is satisfied.

Typical stopping rules use changes in

$$\|\theta^{(t+1)} - \theta^{(t)}\|, \qquad \ell(\theta^{(t+1)}) - \ell(\theta^{(t)}),$$

or both.

EM as a feedback loop



The key question

We have defined an algorithm, but why should it help?

If we maximize $Q(\theta \mid \theta^{(t)})$, why should the *observed-data* log-likelihood $\ell(\theta; Y)$ increase?

The answer comes from a lower-bound argument based on Jensen's inequality.

We will use the discrete latent-variable case for clarity; the same idea extends to integrals.

Jensen's Inequality

Because $\log x$ is concave,

$$\log \mathbb{E}(X) \geq \mathbb{E}(\log X)$$

for a positive random variable X.

Equivalently, for probabilities r_z with $\sum_z r_z = 1$,

$$\log\left(\sum_z r_z a_z\right) \geq \sum_z r_z \log a_z.$$

Jensen lets us replace a difficult log of a weighted sum by a weighted average of simpler log terms — at the cost of obtaining a lower bound.

Constructing the EM lower bound

Let

$$r_z^{(t)} = \mathbb{P}(Z = z \mid Y = y, \theta^{(t)}).$$

Then

$$\begin{aligned}
\ell(\theta; y) = \log \sum_z p(y, z \mid \theta) &= \log \sum_z r_z^{(t)} \frac{p(y, z \mid \theta)}{r_z^{(t)}} \\
&\geq \sum_z r_z^{(t)} \log \frac{p(y, z \mid \theta)}{r_z^{(t)}} \quad \text{(using Jensen)} \\
&= \sum_z r_z^{(t)} \log p(y, z \mid \theta) - \sum_z r_z^{(t)} \log r_z^{(t)}. \\
&= Q(\theta \mid \theta^{(t)}) - [\text{a quantity not depending on } \theta].
\end{aligned}$$

Therefore maximizing Q maximizes a lower bound to the observed-data log-likelihood, with the bound fixed during that M-step.

Theorem: Ascent Property of EM Algorithm

Theorem (Ascent Property of EM Algorithm)

Suppose the conditional distribution of $Z \mid Y = y, \theta^{(t)}$ is well-defined and the M-step is exact, that is,

$$Q(\theta^{(t+1)} \mid \theta^{(t)}) \geq Q(\theta^{(t)} \mid \theta^{(t)}).$$

Then the observed-data log-likelihood is nondecreasing:

$$\ell(\theta^{(t+1)}; y) \geq \ell(\theta^{(t)}; y).$$

This ascent property is the usual monotonicity statement for EM: the observed-data likelihood does not decrease from one exact EM iteration to the next.

See McLachlan and Krishnan (2008), Ch. 3, section 3.2.

Why is the bound tight at the current iterate?

At $\theta = \theta^{(t)}$,

$$\frac{p(y, z \mid \theta^{(t)})}{r_z^{(t)}} = \frac{p(y, z \mid \theta^{(t)})}{p(z \mid y, \theta^{(t)})} = p(y \mid \theta^{(t)}),$$

which does not depend on z.

Therefore Jensen's inequality becomes an equality at the current parameter value.

The lower bound touches the log-likelihood at $\theta^{(t)}$.

This "touch and improve" picture is closely related to the MM principle that we will study later.

Convergence Statement I

Theorem (Convergence of likelihood values)

Under standard regularity conditions, if the observed-data likelihood is bounded above and exact EM iterations are performed, then the sequence

$$\ell(\theta^{(0)}; y),\ \ell(\theta^{(1)}; y),\ \ell(\theta^{(2)}; y), \ldots$$

is nondecreasing and converges to a finite limit.

The ascent property gives monotonicity; boundedness of the likelihood turns monotonicity into convergence of the likelihood values.

Compare Section 3.4 of McLachlan and Krishnan (2008); see also Wu (1983).

Convergence Statement II

Theorem (Limit points are stationary, informally stated)

Under suitable smoothness, compactness, and identifiability-type conditions, every limit point of an EM sequence is a stationary point of the observed-data likelihood. That is, if $\theta^{(t_j)} \to \theta^*$, then typically

$$\nabla \ell(\theta^*; y) = 0,$$

or the appropriate constrained/boundary version of this condition holds.

Stationary point does not mean global maximum. It may be a local maximum, saddle point, or boundary point in irregular problems.

See Section 3.5 of McLachlan and Krishnan (2008); Wu (1983).

What monotonicity does not guarantee

Likelihood ascent is valuable, but it is not the same as global optimality.

- EM can converge to a local maximum.
- It can converge to a saddle point or boundary point in irregular problems.
- Different starting values can lead to different solutions.
- Monotonicity says nothing about speed; EM can be slow near the optimum.

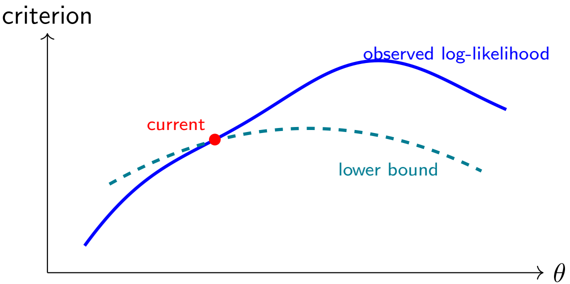
Convergence theory: Wu (1983); McLachlan and Krishnan (2008), Ch. 3.

A geometric picture of EM

Imagine the observed log-likelihood as a difficult curve.

At iteration t:

1. E-step constructs a tractable lower bound touching the likelihood at $\theta^{(t)}$.
2. M-step moves to the maximizer of that bound.
3. Repeating the construction creates a sequence of nondecreasing likelihood values.



An example: Poisson data with missing counts

Suppose

$$X_1, \ldots, X_n \overset{iid}{\sim} \mathrm{Poisson}(\lambda),$$

but only the first r counts are observed.

The remaining $m = n - r$ counts are missing.

Let

$$S_{\mathrm{obs}} = \sum_{i=1}^{r} X_i.$$

> How would EM estimate λ if we deliberately treat the unobserved counts as missing data?

This example is intentionally simple: it lets us see every EM step algebraically.

If all n counts were observed,

$$\ell_c(\lambda) = \sum_{i=1}^{n} \left[X_i \log \lambda - \lambda - \log(X_i!) \right].$$

Ignoring terms that do not involve λ,

$$\ell_c(\lambda) = \left(\sum_{i=1}^{n} X_i \right) \log \lambda - n\lambda + C.$$

The complete-data MLE is therefore

$$\widehat{\lambda}_{\mathrm{comp}} = \frac{1}{n} \sum_{i=1}^{n} X_i.$$

The only missing complete-data sufficient statistic is the sum of the missing counts.

E-step for the Poisson example

At iteration t, each missing count has conditional expectation

$$\mathbb{E}[X_i \mid X_{\mathrm{obs}}, \lambda^{(t)}] = \lambda^{(t)},$$

because the counts are independent.

Hence the expected missing total is

$$\mathbb{E}\left[\sum_{i=r+1}^{n} X_i \middle| X_{\mathrm{obs}}, \lambda^{(t)} \right] = m\lambda^{(t)}.$$

So the E-step replaces the missing sufficient statistic by

$$S_{\mathrm{mis}}^{(t)} = m\lambda^{(t)}.$$

M-step for the Poisson example

The complete-data MLE formula was the full sample mean.

Replace the missing total by its E-step expectation:

$$\lambda^{(t+1)} = \frac{S_{\text{obs}} + m\lambda^{(t)}}{n}.$$

Thus EM produces the update

$$\boxed{\lambda^{(t+1)} = \frac{S_{\text{obs}}}{n} + \frac{m}{n}\lambda^{(t)}}.$$

Numerical iteration

Suppose

$$n = 10, \qquad r = 6, \qquad m = 4, \qquad S_{\text{obs}} = 18.$$

Start with $\lambda^{(0)} = 5$.

Then

$$\lambda^{(1)} = \frac{18 + 4(5)}{10} = 3.8,$$

$$\lambda^{(2)} = \frac{18 + 4(3.8)}{10} = 3.32,$$

$$\lambda^{(3)} = \frac{18 + 4(3.32)}{10} = 3.128.$$

The sequence keeps moving toward 3.

What is the limiting estimate?

At a fixed point λ^*,

$$\lambda^* = \frac{S_{\text{obs}} + m\lambda^*}{n}.$$

Therefore

$$(n - m)\lambda^* = S_{\text{obs}},$$

so

$$\boxed{\lambda^* = \frac{S_{\text{obs}}}{r}}.$$

That is simply the mean of the observed Poisson counts.

The missing counts contain no additional information about λ here, so EM converges to the observed-data MLE.

An useful strategy

This problem is not where EM is needed computationally; direct maximization is easier.

But it cleanly reveals the architecture:

1. Identify missing sufficient statistics.
2. Compute their conditional expectations under current parameters.
3. Plug those expectations into the complete-data estimation formula.
4. Iterate.

In more interesting models, exactly the same logic survives even when the algebra becomes much richer.

Choosing starting values

The starting value $\theta^{(0)}$ matters because the likelihood may have multiple stationary points.

Practical strategies include:

- method-of-moments or simple preliminary estimates;
- random starts;
- clustering-based initialization for mixture models;
- several dispersed starting values, retaining the best final likelihood.

A single EM run from a single arbitrary start can create false confidence.

Monitoring convergence

Useful diagnostics include:

- plot $\ell(\theta^{(t)})$ against iteration;
- monitor $\|\theta^{(t+1)} - \theta^{(t)}\|$;
- inspect whether different starts converge to the same solution;
- check for parameters approaching boundaries or degeneracy.

A common rule is

$$\frac{|\ell^{(t+1)} - \ell^{(t)}|}{1 + |\ell^{(t)}|} < \varepsilon.$$

Common misconceptions

1. **“E-step means replacing each missing value by its mean.”**
   Not always. It means taking the conditional expectation of the *complete-data log-likelihood* or the sufficient quantities appearing in it.
2. **“EM maximizes the complete-data likelihood.”**
   No. Its target is the observed-data likelihood.
3. **“Monotone likelihood means global maximum.”**
   No. Local maxima remain possible.
4. **“EM automatically gives standard errors.”**
   No. Additional calculations are needed for uncertainty quantification.

Some exercises

For each statement, decide whether it is correct.

1. The E-step chooses the most likely value of every missing quantity.
2. The M-step maximizes $Q(\theta \mid \theta^{(t)})$, not necessarily the observed log-likelihood directly.
3. An exact EM iteration can decrease the observed-data likelihood slightly because of numerical noise in theory.
4. Different starting values may lead to different EM solutions.
5. EM is useful only when literal database entries are missing.

Try to justify each answer in one sentence.

Answers

1. **False.** The E-step averages over the conditional distribution of missing/latent quantities; it does not generally make hard assignments.
2. **True.** The M-step optimizes the expected complete-data log-likelihood Q, which is constructed to improve the observed likelihood.
3. **False in exact theory.** Exact EM has nondecreasing observed likelihood; numerical implementations can show tiny violations due to approximation or roundoff.
4. **True.** Nonconcave likelihoods may have multiple local maxima or stationary points.
5. **False.** Latent labels, hidden states, random effects, and other unobserved quantities also create EM problems.

Key Takeaway Messages

1. EM is designed for likelihood problems with missing or latent information.
2. The observed-data likelihood is obtained by summing/integrating over unobserved quantities and may be harder to optimize than the complete-data likelihood.
3. The E-step computes

$$Q(\theta \mid \theta^{(t)}) = \mathbb{E}_{\theta^{(t)}}[\ell_c(\theta; Y, Z) \mid Y].$$

4. The M-step maximizes Q to obtain $\theta^{(t+1)}$.
5. Jensen's inequality explains the lower-bound construction and the nondecreasing likelihood property.
6. Monotone ascent does not guarantee a global maximum or fast convergence.
7. EM is best understood as exploiting a simpler complete-data problem to solve a harder observed-data optimization problem.

Suggested reading after Lecture IV

For first-principles understanding:

- McLachlan and Krishnan (2008), Chapters 1–3: systematic treatment of EM, likelihood ascent, convergence, and examples.
- Little and Rubin (2019), Chapter 8: likelihood methods with missing data and the EM algorithm.
- Dempster et al. (1977): the foundational EM paper; especially the general framework and monotonicity idea.

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